1.Develop a program to blink 5 LEDs back and forth.

```
// CONNECT CN4 TO CN9
int leds[] = {15,12,3,5,19}; // Assign pins for the LEDs
int numLEDs = 5;
int delayTime = 1000; // Delay between LED changes in milliseconds
int direction = 1; // 1 for forward, -1 for backward
int currentLED = 0;

void setup() {
  for (int i = 0; i < numLEDs; i++) {
    pinMode(leds[i], OUTPUT);
  }
}

void loop() {
  // Turn off all LEDs
  for (int i = 0; i < numLEDs; i++) {
    digitalWrite(leds[i], LOW);
```

```
  }

  // Turn on the current LED
  digitalWrite(leds[currentLED], HIGH);
  delay(delayTime);

  // Update the current LED index, reversing
direction if needed
  currentLED += direction;
  if (currentLED >= numLEDs - 1 ||
currentLED <= 0) {
    direction *= -1; // Reverse direction
  }
}
```

2. Develop a program to interface a relay with Arduino board.

```
//Define pin
// RM17 - RM9 connected
int buzzer_pin = 8;     // Arduino pin #23/D9
 or Board pin P38

void setup()
```

```
{
  pinMode(buzzer_pin, OUTPUT);
  Serial.begin(9600);
  digitalWrite(buzzer_pin,HIGH);
}

void loop() {
  digitalWrite(buzzer_pin, LOW);
  delay(1000);
  digitalWrite(buzzer_pin, HIGH);
  delay(1000);
}
```

3. Develop a program to deploy an intrusion detection system using Ultrasonic and sound sensors.

```
//Security Alarm with Ultrasonic Sensor//
//Connect RM 21 To RM4 Ultrasonic sensor
//Coneect RM17 To RM9 Buzzer
#define trigPin 6
#define   echoPin 7
#define buzzer 9
float new_delay;
```

```
void setup()
{
   Serial.begin (9600);
  pinMode(trigPin, OUTPUT);
  pinMode(echoPin, INPUT);
   pinMode(buzzer,OUTPUT);

}

void loop()
{
  long duration,   distance;
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
   digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin,   LOW);
  duration = pulseIn(echoPin, HIGH);
  distance = (duration/2) / 29.1;
   new_delay= (distance *3) +30;
  Serial.print(distance);
  Serial.println("   cm");
  if (distance < 100)
```

```
  {
   digitalWrite(buzzer,HIGH);
   delay(new_delay);
    digitalWrite(buzzer,LOW);

  }
  else
  {
    digitalWrite(buzzer,LOW);

   }

 delay(200);
}
```

4. Develop a program to control a DC motor with Arduino board

```
//  Make the connection
//CONNECT RM17 - RM9
//USE MALE-MALE & CONNECT DC MOTOR TO CN11
```

```
// USE MALE-FEMALE & CONNECT DC MOTOR TO VCC
// USE MALE-FEMALE & CONNECT CN11 TO GND


int Relay_pin = 8;     // Arduino pin #23/D9  or Board pin P38
void setup()
{
  pinMode(Relay_pin, OUTPUT);  // Declare relay pin as OUTPUT
  Serial.begin(9600);           // Start the Serial communication
  digitalWrite(Relay_pin,HIGH);  // Make the output as high
}
void loop() {
  digitalWrite(Relay_pin, LOW);  //Make the pin low
  delay(1000);                  // delay of 1 sec
```

```
  digitalWrite(Relay_pin, HIGH);  // make the pin as high
  delay(1000);                    // delay of 1 sec and continue in the loop
}
```

5. Develop a program to deploy smart street light system using LDR sensor.

```
//CONNECT RM20 TO RM4
//CONNECT CN4 TO CN10

int res = 0;       /* declaring the variable that will store the value of photoresistor*/
int sensor = A0;     /* assigning Arduino pin for photoresistor*/
int led= 5;  /* assigning Arduino pin for LED*/
void setup() {
Serial.begin(9600);      /*setting the baud rate for serial communication*/
```

```
pinMode(5, OUTPUT);     /* assigning mode to LED pin */
            }
void loop() {               // put your main code here, to run repeatedly:
res = analogRead(sensor);           /* getting the value of photoresistor*/
Serial.println(res);            /* displaying the photoresistor value on serial monitor */
   if(res > 100) {         /* when the value of sensor is less than 100 */
Serial.println(" Low intensity ");
digitalWrite(5,LOW);            /* keep the LED off*/
                }
   else {           /* otherwise turn the light on */
Serial.println("High Intensity ");
digitalWrite(5,HIGH);           /* turn the LED on*/
        }
```

```
delay(1000);
}
```

6. Develop a program to classify dry and wet waste with the Moisture sensor (DHT22).

```
//Define pin
// CONNECT MOISTURE SENSOR TO SOIL MOISTURE SENSOR ON BOARD
// RM6 - RM23 connected
int soil_pin = 1;                // Arduino pin #9/A1  or Board pin P11

void setup() {
  pinMode(soil_pin, INPUT);
  Serial.begin(9600);
}
```

```
void loop() {
  int moisture_value = analogRead(soil_pin);
  Serial.print("WET Moisture value : ");
  Serial.println(moisture_value);
  // Checks if it has reached the threshold value

  delay(1000);
}
```

7. Develop a program to read the pH value of a various substances like milk, lime and water.

8. Develop a program to detect the gas leakage in the surrounding environment.

```
/*
*  Example testing sketch for Analog Gas sensor
*
```

```
* //connect RM5 To RM22 and see the values in serial window
*
*/
//Define pin
// RM5 - RM22 connected
int gas_pin = 0;     // Arduino pin #8/A0  or Board pin P11

void setup() {
  pinMode(gas_pin, INPUT);
  Serial.begin(9600);
}

void loop() {
  int gas_value = analogRead(gas_pin);

  Serial.print("Gas Value : ");
  Serial.println(gas_value);
```

```
  // Checks if it has reached the threshold value


  delay(1000);}
```

9. Develop a program to demonstrate weather station readings using Arduino.

```
/*Short RM19 & RM2 */
#include <DHT.h>

// Define the DHT sensor type and the pin it's connected to
#define DHTPIN 4      // Pin which is connected to the DHT sensor
#define DHTTYPE DHT22 // DHT22 (AM2302)
// Initialize DHT sensor
DHT dht(DHTPIN, DHTTYPE);
void setup() {
```

```
  Serial.begin(9600);
  dht.begin(); // Initialize the sensor
}
void loop() {
  // Wait a few seconds between measurements.
  delay(2000);
  // Reading temperature or humidity takes about 250 milliseconds!
  float humidity = dht.readHumidity();
  float temperature = dht.readTemperature(); // Default is Celsius
  // Check if any reads failed and exit early (to try again).
  if (isnan(humidity) || isnan(temperature)) {
    Serial.println("Failed to read from DHT sensor!");
    return;
  }
```

```
  Serial.print("Temperature = ");
  Serial.print(temperature);
  Serial.println(" *C");
  Serial.print("Humidity = ");
  Serial.print(humidity);
  Serial.println(" %");
}
```

10. Develop a program to setup a UART protocol and pass a string through the protocol.

```
/*  Example sketch to test Bluetooth
** Xtrans Solutions Pvt Ltd
* www.xtranssolutions.com
* Connect RM24 to RM25
* to check PIN use Command AT+PIN
* * Now you can send and receive messages from any serial application like Hyper
```

```
Terminal to Aurduino Serial Monitor and vice versa
*/
#include<SoftwareSerial.h>
/* Create object named bt of the class SoftwareSerial */
SoftwareSerial bt(0,1); /* (Rx,Tx) */
void setup() {
  bt.begin(115200); /* Define baud rate for software serial communication */
  Serial.begin(115200); /* Define baud rate for serial communication */
}
void loop() {
    if (bt.available()) /* If data is available on serial port */
    {
     Serial.write(bt.read()); /* Print character received on to the serial monitor */
    }
```

```
    if(Serial.available())
    {
      bt.write(Serial.read());
    }
}
```

11. Develop a water level depth detection system using Ultrasonic sensor.

12. Develop a program to simulate interfacing with the keypad module to record the keystrokes.

```
//install library Keypad by Mark Stanley, Alexander Brevig

#include <Key.h>

#include <Keypad.h>

//C0NNECT  CN5 TO CN9

/*4x4 Matrix Keypad connected to Arduino

This code prints the key pressed on the keypad to the serial port*/
```

```
//Go To sketch ---> Include Library --> Manage Library --->Search for Keypad library and install


const byte numRows= 4; //number of rows on the keypad
const byte numCols= 4; //number of columns on the keypad


//keymap defines the key pressed according to the row and columns just as appears on the keypad
char keymap[numRows][numCols]=
{
{'0', '4', '8', 'C'},
{'1', '5', '9', 'D'},
{'2', '6', 'A', 'E'},
{'3', '7', 'B', 'F'}
};
```

```
//Code that shows the the keypad connections to the arduino terminals
// CN9-CN5 connected
byte colPins[numRows] = {10,11,12,13}; //Rows 0 to 3
byte rowPins[numCols]= {A2,A3,A4,A5}; //Columns 0 to 3

//initializes an instance of the Keypad class
Keypad myKeypad= Keypad(makeKeymap(keymap), rowPins, colPins, numRows, numCols);

void setup()
{
Serial.begin(9600);
}

//If key is pressed, this key is stored in 'keypressed' variable
```

```
//If key is not equal to 'NO_KEY', then this key is printed out
//if count=17, then count is reset back to 0 (this means no key is pressed during the whole keypad scan process
void loop()
{
char keypressed = myKeypad.getKey();
if (keypressed != NO_KEY)
{
Serial.print(keypressed);
}
}
```